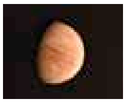
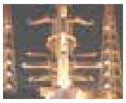


Smallest exoplanet discovered
Researchers discovered an exoplanet which is less than half the size of the Earth.
https://dgit.in/sep21-33



Space is hard
Indian rocket suffers catastrophic failure during launch, Earth-watching satellite lost.
https://dgit.in/sep21-34

lab in Canada, using liquid nitrogen to reduce the temperature to minus 50 degrees Celsius, which is similar to the temperature found at the south pole of Mars. These samples turned out to be nearly perfect matches to the radar observations. The researchers then looked through the NASA's Mars Reconnaissance Orbiter data and found smectites in the regions around the areas flagged as potential underground lakes. The findings do not rule out the possibility that these are underground lakes, but offer a better explanation for the radar findings.

Jeffrey Plaut of NASA's Jet Propulsion Laboratory explains, "In planetary science, we often are just inching our way closer to the truth. The original paper didn't prove it was water, and these new papers don't prove it isn't. But we try to narrow down the possibilities as much as possible in order to reach consensus."

LIFE IN THE ICE MOONS

The ice moons of the gas giants in the solar system have global subsurface oceans, where tidal heating from the planets and other geological processes on the moon can provide sufficient energy for life to exist, given the right conditions.

Enceladus

Credit: NASA

Data captured by Cassini as it flew through the plumes of Enceladus indicate an unknown source of methane on Saturn's moon, which could be life. Titan, another moon of Saturn, also has the right conditions to harbour life based on two different types of biochemistry.

Europa

Credit: NASA/JPL-Caltech

NASA has scheduled the Europa Clipper mission for an October 2024 launch, to create a global surface map of the Jovian moon, measure the subsurface ocean, identify underground lakes below the ocean, all with the objective of looking for signs of past or present life.

PERSEVERANCE ON THE HUNT

On 18 February 2021, the Perseverance rover with the Ingenuity Mars helicopter on board executed a soft landing at the Jezero crater in Mars. The landing was pretty similar to the one made by Curiosity a decade ago, with a sky crane that used reverse thrusters to hover over the planet while the rover was gently lowered to the surface. The landing site was at the Jezero crater, where 3.5 billion years ago a river delta emptied into a lake. Signs of any ancient microbial life can be picked up with careful study of the rocks in the region, and that is the mission of Perseverance.

Credit: NASA/JPL-Caltech/MSSS

The image of a rock known as "Foux" captured by the Watson camera on the robotic arm of Perseverance.

On board the robotic arm of the rover are a pair of instruments known as Sherloc and Watson. Sherloc stands for Scanning Habitable Environments with Raman & Luminescence for Organics & Chemicals.while Watson stands for Wide Angle Topographic Sensor for Operations and eNgineering. A purple (technically ultraviolet) laser is part of Sherloc's setup, which uses the principle of Raman spectroscopy to shine a light into the rocks and understand their molecular composition by checking how the light is scattered. The Watson instrument takes high resolution images of the rocks, allowing scientists to figure out grain size and texture. The data from Sherloc can be superimposed on the data from Watson, to allow researchers to understand how minerals got deposited over time and how the layers overlap. Combined with the data from another instrument, known as Planetary Instrument for X-ray Lithochemistry (PIXL), the robotic arm can look for past signs of microbial life in the rocks such as a clump of organics in one side of a rock. Sherloc and Watson could together be responsible for finding the first traces of life beyond Earth. d